

# Template and Tutorial for an Analysis

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## Introduction

This is a template to use for qmd reports

These are useful for a variety of reasons. It allows you to access your analyses, figures, and tables all in one pdf/html file without having to run everything every time you boot up your computer. It also allows you to organize your thoughts within the project so everyone can catch up easily. This is still useful in individual projects when you want to show someone what you have done without forcing them to run your code on their machine or you constantly updating a google document (I have made this mistake in the past)

I am assuming you have some R experience in doing analyses, but are not as comfortable formatting a report. I am also assuming that you have some in-class experience in doing a project, but have never done something without much direction. Open-ended projects are exciting, but it's easy to fall into rabbit holes or forgo the important aspect of EDA.

The rest of the document is structured as follows:

- 1) Formatting, LaTeX, showing relevant bits of code
- 2) Exploratory Data Analysis + gtsummary
- 3) Thinking about a research question and how to approach it
- 4) Present your science!

## Formatting, LaTeX, showing relevant bits of code

First off, let's discuss formatting in a qmd. A report document is all the relevant bits and pieces you want to share to your reader. Sometimes, showing code is important. Oftentimes, it's not necessary. However, you need all your code within the outline (more advanced, you can move functions to a separate document and call it from here, but I won't talk about that). The current settings for this document are *echo: false* (do not show the code, but run it), *warning: false* (suppress any warning messages), *error: false* (suppress any error messages: i.e. `error in exists(cacheKey...)`), and *message: false*. This means all chunks will have these default settings unless otherwise specified.

Did you see that? Well if you're just reading the report pdf, then you didn't see anything. If you're in Rstudio or equivalent, you'll see that I just called the *mtcars* dataset and the tidyverse library

```
library(ggplot2)
summary(a)
```

```

      mpg
Min.   :10.40
1st Qu.:15.43
Median :19.20
Mean   :20.09
3rd Qu.:22.80
Max.   :33.90

```

To not run a chunk at all (there are many reasons for this, but I am not going to list them out), use *eval: false* as follows.

```
2 + 3
```

I sneakily added an echo: true statement so you can see that the code shows, but nothing runs. For example, introducing a function to the reader is a good idea

## LaTeX

Notice that I just created a subheading. The more #’s you put, the more sub it gets. You’ll also see in the table of contents that we got a clean subsection

If you want to use LaTeX, you can do so directly in R like so (as a bonus, it shows up in Rstudio)

$$\mathbb{E}(X) = \int_{\Omega} X d\mathbb{P}$$

## Exploratory Data Analysis

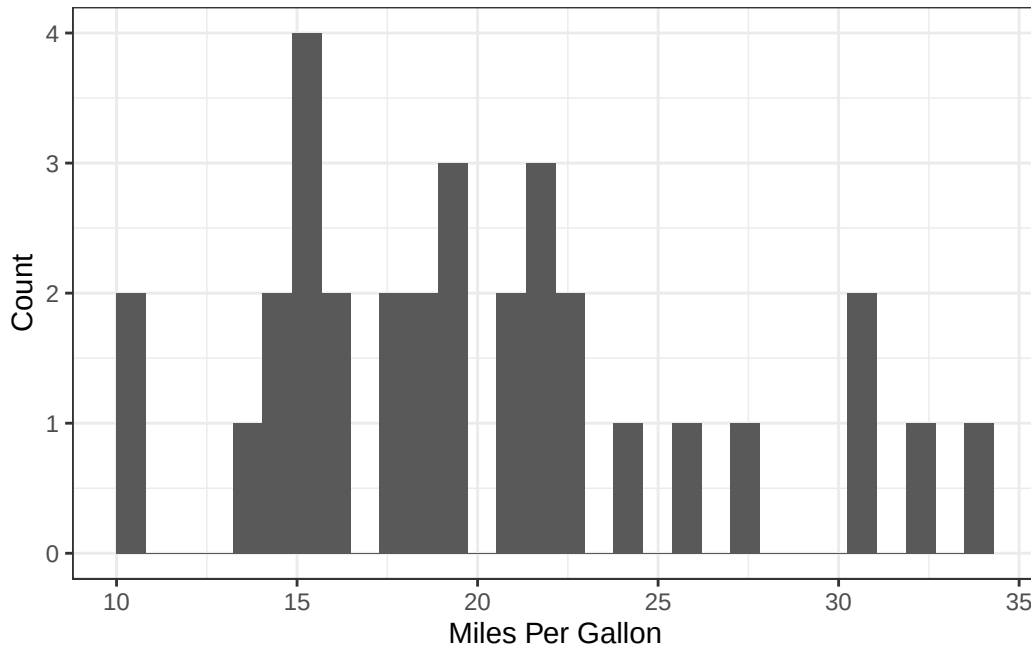
Before beginning any analysis, you need to spend a lot of time thinking about what variables I have and what do they look like. Of course it’s easy to do `summary()`, `head()` or `names()`. These are all great starting points, but we should be a bit more comprehensive, especially if we want to see what questions we can answer with the data. I will be using the *mtcars* dataset as a toy example

```

[1] "mpg"  "cyl"  "disp" "hp"    "drat" "wt"    "qsec" "vs"    "am"    "gear"
[11] "carb"

```

If we want to focus on a continuous variable, maybe a histogram is a good idea to check for outliers and to understand what the data is like



Maybe we want to understand how different *cyl* values affect mpg? The point of this graph is to show that things begin to get smushed when figures get too large. When this happens, there are ways to enlarge the figures (google this), but too much information is not useful when it's hard to understand what's going on. Additionally, to create more of these types of visualizations, you will have to remember how to pivot your data

A very important part of EDA, especially for a project like advancement, is understanding who is in your dataset. You need to create tables to summarize who is in your data, even if you don't end up using this information in the future. For example, what is the distribution of everyone's majors by class year? Should we double/triple/quadruple count double/triple/quadruple majors?

I like using the *gtsummary* package, which makes nice *gt* tables automatically for you. Note that the reason why I type *dplyr::select* for example is that it specifies which package the select function I'm running should come out of.

The *gtsummary* package is extremely powerful, as it can create tables for all types of regression outputs, these summary tables, and even saving these tables as png's to put in a presentation. For more information, visit: <https://www.danielsjoberg.com/gtsummary/> (MSK represent).

As an aside, the formatting of these reports gets really messy. It's honestly fine as long as you quote your tables, such as Table 1 and Figure 1

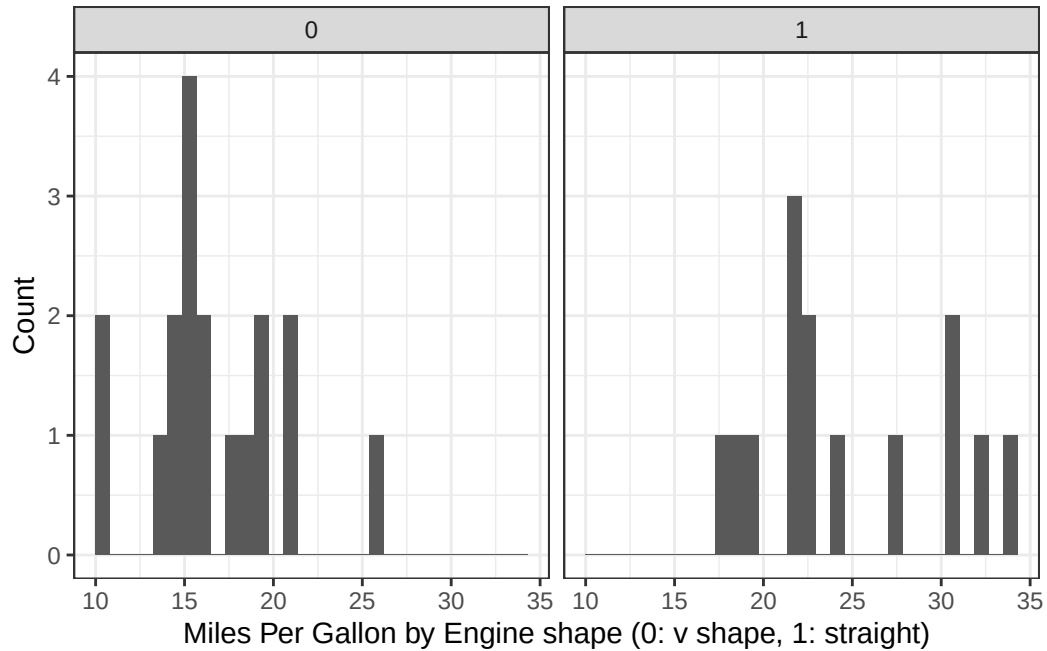


Figure 1: This is how you add a caption to your figure

Table 1: This is how you add a caption to your table.

| Characteristic | 0 N = 18 <sup>†</sup> | 1 N = 14 <sup>†</sup> |
|----------------|-----------------------|-----------------------|
| mpg            | 15.7 (14.7, 19.2)     | 22.8 (21.4, 30.4)     |
| disp           | 311 (276, 360)        | 121 (79, 168)         |

<sup>†</sup>Median (Q1, Q3)

```
library(gtsummary)

mtcars |>
  dplyr::select(vs, mpg, disp) |> #only select the variables you want to look at
  gtsummary::tbl_summary(by = vs) #this is just like the summarize function!
```

Of course the type of EDA can become much more complicated. Maybe we have a 5x5 matrix for a multivariate look into the data. Maybe we have small multiples (these are cool). Maybe we have a kaplan-meier curve if we're working with time-to-event data. The possibilities are endless, but the main goal is for us and the audience to understand what's going on.

*how do i add "figure 1, and the description... I did it for 495*

## Thinking about Analysis and Revising

Now that we understand our variables, we move onto how we formulate a question and how we answer it. I will spill out some tips for you

- A more complicated analysis is not always necessary unless you realize you have limitations in the methods you're currently using.
- When you identify a new method, make sure you do the work to ensure it's appropriate for the data. It doesn't have to be perfect (no model/analysis ever is), but it should meet the conditions so everything can be interpreted.
- Document your process in a separate document so you don't mess up the master document. When you're ready to add something, copy-paste it.

## Conclusion

You got this!